

M15 / ASSESSMENT

M15 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

Contents

Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

Read this first 3

M15 | CLOSED-BOOK MODULE REVISION TEST 3

Read this first

M15 | CLOSED-BOOK MODULE REVISION TEST

No forced timer. Close learner/review/answers. Save the genuine attempt. Open protected answers only after you have finished.

1. DE15-01 - Cloud fundamentals for data engineers

Draw a six-box architecture for orders arriving from an operational database into a Fabric lakehouse and then a gold table. Label storage, compute, identity and monitoring boundaries. Do not use more than two Fabric compute item types.

2. DE15-02 - Identity, IAM and least privilege

Design roles for four personas: developer, data consumer, deployment automation and production support. State the minimum workspace-level role and one thing each persona must not be allowed to do.

3. DE15-03 - Azure Storage concepts and OneLake

Design a Files/Tables layout for raw CRM JSON, cleaned customer Delta data and a curated customer_daily table. State why each location is chosen.

4. DE15-04 - Fabric workspace and item model

Create a proposed DEV workspace inventory for an order pipeline with at least one lakehouse, pipeline, notebook and environment. Draw dependency arrows.

5. DE15-05 - OneLake paths, files, tables and shortcuts

For three datasets - daily exchange rates, immutable regulatory snapshot, and a 4-TB shared customer dimension - choose shortcut or copy and defend the decision.

6. DE15-06 - Fabric Lakehouse and Delta tables

Design Files and Tables areas for orders, customers and daily_sales. State which objects Spark writes and which are queried from the SQL endpoint.

7. DE15-07 - Fabric notebooks

Refactor a pseudo-notebook with hard-coded source path, secret and run date into parameterized sections. Write the validation checks that must happen before any write.

8. DE15-08 - Spark in Fabric

Write a runtime evidence checklist for DEV and PROD. Include at least Spark, Python, Delta/runtime, attached environment and one library version your project depends on.

9. DE15-09 - Fabric Data Pipelines

Design a pipeline for daily customer changes with source lookup, copy, transform, validate and commit. Mark which activities are safe to retry and why.

10. DE15-10 - SQL analytics endpoint and warehouse choice

Classify four needs: ad-hoc SELECT over Delta, Spark MERGE, stored-procedure-based SQL transformation, and Power BI consumption. Choose the appropriate Fabric surface for each.

11. DE15-11 - Medallion architecture in Fabric

Define Bronze/Silver/Gold contracts for ecommerce orders. Include grain, required keys, one quality check and allowed consumers for each layer.

12. DE15-12 - Incremental loads and MERGE in cloud

Design incremental loading for customers with `updated_at`, `customer_id` and `is_deleted`. State selection predicate, MERGE key, delete handling and commit point.

13. DE15-13 - Permissions, connections and secrets

Review five configuration values - workspace name, table name, API base URL, API token, Key Vault secret name. Classify which may live in source control and which must not.

14. DE15-14 - Monitoring, reliability and cost awareness

Create an M15 dashboard spec with six metrics and alert thresholds for an hourly ingestion/transformation pipeline.

15. DE15-15 - Git, deployment and environment separation

List which parts of a lakehouse+notebook+pipeline solution you expect in Git and which should be environment-specific. Add one post-deploy smoke test.

16. DE15-16 - End-to-end Fabric data engineering case

Write a one-page design for the same case before looking back at the worked solution. Include items, layers, identities, progress state, serving surface and monitoring evidence.

17. INT-01 - Runtime boundary

Why is it wrong to force the M13 local Spark/Delta versions into Fabric Runtime 2.0?

18. INT-02 - Control-plane vs data-plane failure

Give one example where a green pipeline status can coexist with wrong business data.

19. INT-03 - Security + deployment

Explain why a notebook may deploy successfully yet fail in PROD with 403.

20. INT-04 - Cost + correctness

Give one optimization that reduces Fabric work without weakening reconciliation or late-data handling.